

```
In [50]: %config SqlMagic.style = '_DEPRECATED_DEFAULT'
```

Load Proper Libraries

```
In [51]: %load_ext sql
```

The sql extension is already loaded. To reload it, use:
%reload_ext sql

Create Database

```
In [52]: %sql sqlite:///housing.db
```

Q1.Return the first 7 records from the housing table

A1.

```
In [53]: %%sql

select * from housing

limit 7

* sqlite:///housing.db
  sqlite:///open6608_database.db
Done.
```

```
Out[53]:
```

	entry	city	year	month	sales	volume	median	listings	inventory
	1	Abilene	2000	1	72	5380000	71400	701	6.3
	2	Abilene	2000	2	98	6505000	58700	746	6.6
	3	Abilene	2000	3	130	9285000	58100	784	6.8
	4	Abilene	2000	4	98	9730000	68600	785	6.9
	5	Abilene	2000	5	141	10590000	67300	794	6.8
	6	Abilene	2000	6	156	13910000	66900	780	6.6
	7	Abilene	2000	7	152	12635000	73500	742	6.2

Q2. How many records in the data had sales greater than 100 and a median sales price greater than 100000? Write a query to return the result.

A2.

```
In [54]: %%sql
```

```
select count(*)  
  
from housing  
  
where sales > 100  
and median > 100000
```

```
* sqlite:///housing.db  
  sqlite:///opan6608_database.db  
Done.
```

```
Out[54]: count(*)  
         5106
```

Q3. Show the number of records in the data that had sales greater than 100 or a median sales price less than 100000 by month. The result should contain two columns: month and num_records.

A3.

```
In [55]: %%sql  
  
select  
month,  
count(*) as num_records  
  
from housing  
  
where sales > 100  
and median > 100000  
  
group by month
```

```
* sqlite:///housing.db  
  sqlite:///opan6608_database.db  
Done.
```

Out[55]:

month	num_records
1	351
2	389
3	428
4	443
5	470
6	488
7	485
8	448
9	407
10	406
11	392
12	399

month	num_records
1	351
2	389
3	428
4	443
5	470
6	488
7	485
8	448
9	407
10	406
11	392
12	399

Q4. Return a result that shows the mean annual sales by city and year, excluding cities that begin with the letter H. The result should be in descending order by mean sales, limited to 20 records, and should contain three columns: city, year, and mean_annual_sales

A4.

```
In [56]: %%sql

select
    city,
    year,
    round(avg(sales), 2) AS mean_annual_sales

from housing

where
    city not like 'H%'

group by
    city,
    year

order by
    mean_annual_sales DESC

limit 20
```

```
* sqlite:///housing.db
  sqlite:///open6608_database.db
Done.
```

Out[56]:

city	year	mean_annual_sales
------	------	-------------------

Dallas	2006	5352.17
Dallas	2015	5247.86
Dallas	2014	5012.83
Dallas	2005	4998.33
Dallas	2013	4976.17
Dallas	2007	4974.58
Dallas	2004	4542.83
Dallas	2008	4237.33
Dallas	2012	4200.92
Dallas	2003	4106.5
Dallas	2002	3933.25
Dallas	2001	3916.0
Dallas	2009	3824.25
Dallas	2000	3787.17
Dallas	2011	3554.67
Dallas	2010	3531.92
Austin	2015	2696.86
Austin	2014	2579.42
Austin	2013	2536.33
Austin	2006	2523.67

Q5. How many sales were there in Austin in December 2010? Return the city, year, and sales in the result.

A5.

```
In [57]: %%sql

select sales
from housing
```

```
where city = 'Austin'  
and month = 12  
and year = 2010
```

```
* sqlite:///housing.db  
  sqlite:///opan6608_database.db
```

Done.

Out [57]: **sales**

1561

Q6. In 2010 in Austin, which month had the most sales? Return a result that shows the number of sales in Austin in 2010 by month.

A6.

In [58]: **%%sql**

```
select month, max(sales)  
  
from housing  
  
where city = 'Austin'  
and year = 2010
```

```
* sqlite:///housing.db  
  sqlite:///opan6608_database.db
```

Done.

Out [58]: **month max(sales)**

5	2286
---	------

Q7. In which city-month combinations did the greatest numbers of sales, outside of Austin? Return a result that contains the top five cities' sales and the month it took place in 2010, excluding Austin.

A7.

In [59]: **%%sql**

```
select  
    city,  
    month,  
    sales  
  
from housing  
  
where  
    city <> 'Austin'  
and
```

```

    year = 2010

order by sales desc

limit 5

```

```

* sqlite:///housing.db
  sqlite:///opan6608_database.db
Done.

```

```

Out[59]:

```

city	month	sales
Houston	5	6335
Houston	6	6117
Houston	4	5932
Houston	3	5389
Dallas	5	4891

Q8. In which city, year, and month did the fewest sales take place?
The result should return four columns: fewest_sales, city, year, and month

A8.

```

In [60]: %%sql

select
    min(sales) as fewest_sales,
    city,
    year,
    month

from housing

```

```

* sqlite:///housing.db
  sqlite:///opan6608_database.db
Done.

```

```

Out[60]:

```

fewest_sales	city	year	month
6	San Marcos	2011	10

Q9. For the entire dataset, what were the mean sales in each year?

A9.

```

In [61]: %%sql

select

```

```
    year,  
    round(avg(sales), 2) as avg_sales  
  
from housing  
  
group by year
```

```
* sqlite:///housing.db  
sqlite:///open6608_database.db
```

Done.

Out[61]:

year	avg_sales
2000	403.05
2001	419.3
2002	425.0
2003	459.98
2004	514.49
2005	572.55
2006	629.95
2007	593.66
2008	501.2
2009	462.42
2010	440.24
2011	446.3
2012	520.02
2013	607.05
2014	626.3
2015	646.35

Q10. Return only the years that had mean sales greater than 600.
Which years does this include?

A10.

In [62]:

```
%%sql
```

```
select year
```

```
from housing
```

```
group by year
```

```
having avg(sales) > 600
```

```
order by year
```

```
* sqlite:///housing.db
```

```
sqlite:///open6608_database.db
```

```
Done.
```

```
Out[62]: year
```

```
2006
```

```
2013
```

```
2014
```

```
2015
```